Surface Area of Pyramids

Lesson 10-5

Name: Date: Class:

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

Like the Great Pyramid of Giza — a square on the bottom, 4 triangles slanting up to a point

Pyramid

A solid with a flat bottom and triangle sides that meet at one point on top.

If you slide your finger from the bottom edge up the triangle face to the top point — that distance is the slant height

Slant height

The height of a side triangle, measured along its slanted face.

A square pyramid has 4 lateral faces — one triangle for each side of the square base

Lateral face

A triangle side of a pyramid, not the bottom.

A square pyramid sits on a square base; base area = side × side

Base

The flat bottom of a pyramid.

The pointy top of a pyramid — all the slanted edges connect here

Apex

The point at the top of a pyramid where the sides meet.

For a square pyramid: lateral area = 4 × (½ × base edge × slant height)

Lateral area

The total area of just the side triangles, not the bottom.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can find the surface area of a pyramid by adding the base area and the lateral faces.

1️⃣ Notice

Your class is building a pyramid-shaped display to showcase the time capsule at the school entrance.

2️⃣ Key idea

I can find the surface area of a pyramid by adding the base area and the lateral faces.

3️⃣ Words I'll use

Pyramid **Slant height** **Lateral face** **Base** **Apex** **Lateral area**

🤔 Think About It

- How many faces does a square pyramid have?
- What shape is the base? What shape are the side faces?
- What is the difference between the height of the pyramid and the slant height?

👀 Watch

Watch your teacher model one example. Jot what you see.

🤝 We try

Solve the next one together as a class.

✏️ You try

Now try one on your own.

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

A square pyramid time capsule display has a square base and four triangular faces that meet at a point on top. To find its surface area, which parts do you need to measure?

Level 1 support

Start here: Surface area of a pyramid is the base area plus the areas of all the triangular lateral faces.

💡 **Need a hint?**

Sentence starters (optional)

I need the area of the ____ plus the areas of the ____ triangular faces.
Necesito el área de la ____ más las áreas de las ____ caras triangulares.

The surface area is the ____ area plus the ____ faces.
El área de superficie es el área de la ____ más las caras ____.

Word bank: pyramid | base | lateral face | slant height | surface area

Listen for: *Students explain a square pyramid's surface area = base area + the four triangular lateral faces, and that you add every face together.*

Level 2

Why do you use the slant height, not the pyramid's vertical height, to find the area of each triangular face?

- I use the slant height because it is the ____ of each triangular face.
- The vertical height would not work because ____.

EXPLORE

For Display A (square base 6×6, slant height 8), how did you combine the base area and the four triangular faces to get the total surface area?

Level 1 support

Start here: Add the base area to four times the area of one triangular face (½ × base × slant height).

💡 **Need a hint?**

Sentence starters (optional)

The base area is ____ × ____ = ____, and one triangular face is ½ × 6 × 8 = ____.
El área de la base es ____ × ____ = ____, y una cara triangular es ½ × 6 × 8 = ____.

The total surface area is the base plus 4 × ____ = ____ in².
El área total de superficie es la base más 4 × ____ = ____ in².

Word bank: pyramid | base | lateral face | slant height | surface area

Listen for: *Students compute base = 36 in², one face = 24 in², four faces = 96 in², total SA = 132 in², distinguishing base from lateral faces.*

Level 2

Two pyramids have the same 8×8 base but different slant heights. How will their surface areas compare, and which part stays the same?

- The pyramid with the larger slant height has ____ surface area because ____.
- The ____ area stays the same because slant height only affects ____.

CONNECT

A museum builds a square pyramid display case with base edge 3 ft and slant height 4 ft, and the base is open (no glass). Talk through how to find how much glass the 4 triangular sides need.

Level 1 support

Start here: Find one triangular face area, multiply by 4, and skip the base since it is open.

💡 **Need a hint?**

Sentence starters (optional)

Each triangular face has area ½ × ___ × ___ = ___ ft².

Cada cara triangular tiene área ½ × ___ × ___ = ___ ft².

Four faces need ___ ft² of glass, and the base needs ___ because it is open.

Cuatro caras necesitan ___ ft² de vidrio, y la base necesita ___ porque está abierta.

Word bank: pyramid | base | lateral face | slant height | surface area

Listen for: *Students compute one face = ½ × 3 × 4 = 6 ft², four faces = 24 ft², and explain the open base is not counted, so only the lateral area matters.*

Level 2

How does leaving the base open change which formula you use, and how would the answer change if the base needed glass too?

- Leaving the base open means I use only the ___ area because ___.
- If the base needed glass, I would add ___ to the total.

REFLECT

A classmate found a square pyramid's surface area by adding only the four triangular faces. What did they forget, and how would you correct it?

Level 1 support

Start here: They left out the base area, which must be added to the lateral faces for total surface area.

💡 **Need a hint?**

Sentence starters (optional)

They forgot to add the ___ area to the triangular faces.
Olvidaron sumar el área de la ___ a las caras triangulares.

The correct total surface area is the base plus the ___ lateral faces.
El área de superficie total correcta es la base más las ___ caras laterales.

Word bank: pyramid | base | lateral face | slant height | surface area

Listen for: *Students identify the missing base area and state that total surface area = base area + all lateral (triangular) faces.*

Level 2

When is it correct to leave out the base (like an open display case), and when must it be included? Generalize a rule.

- You leave out the base when ___, but include it when ___.
- In general, total surface area of a pyramid is always ___.

Guided Examples

Example 1

A square pyramid has a base edge of 5 in and a slant height of 7 in. What is the surface area?

Solution: Base: $5 \times 5 = 25$ in². Each lateral face: $\frac{1}{2} \times 5 \times 7 = 17.5$ in². Four faces: $4 \times 17.5 = 70$ in². SA = $25 + 70 = 95$ in².

Answer: A. 95 in²

Example 2

How many lateral (triangular) faces does a square pyramid have?

Solution: A square pyramid has a square base and 4 triangular lateral faces — one for each side of the square.

Answer: A. 4

Example 3

What is the area of one triangular face with base 8 in and slant height 6 in?

Solution: Area of a triangle = ½ × base × height = $\frac{1}{2} \times 8 \times 6 = 24$ in².

Answer: A. 24 in²

Write About the Math

The Writing Revolution

I can explain my work using the words pyramid, slant height, lateral face, and base.

1. Kernel Sentence

subject + verb

Model: Pyramid is a solid with a flat bottom and triangle sides that meet at one point on top.
Pirámide es un sólido con un fondo plano y lados triangulares que se unen en un punto arriba.

Write a kernel sentence about pyramid. Use a subject and a verb.
Escribe una oración base sobre pirámide. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Pyramid matters in math
Pirámide importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*

Pyramid matters in math because ____.
Pirámide importa en matemáticas porque ____.

____

but *pero*

Pyramid matters in math, but ____.
Pirámide importa en matemáticas, pero ____.

____

so *entonces*

Pyramid matters in math, so ____.
Pirámide importa en matemáticas, entonces ____.

____

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about pyramid.
Di un hecho verdadero sobre pyramid.

Pyramid ____.

Question
Pregunta

Ask a question about pyramid.
Haz una pregunta sobre pyramid.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about pyramid.
Muestra entusiasmo sobre pyramid.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with pyramid.
Dile a un compañero qué hacer con pyramid.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

I need the area of the ____ plus the areas of the ____ triangular faces.
Necesito el área de la ____ más las áreas de las ____ caras triangulares.

The surface area is the ____ area plus the ____ faces.
El área de superficie es el área de la ____ más las caras ____.

The base area is ____ × ____ = ____, and one triangular face is ½ × 6 × 8 = ____.
El área de la base es ____ × ____ = ____, y una cara triangular es ½ × 6 × 8 = ____.

Try It

Solve on your own. Check the answer key when you are done.

1. What is the area of one triangular face with base 8 in and slant height 6 in?

A. 24 in^2

B. 48 in^2

C. 14 in^2

D. 24 in^3

Show your work:

2. A square pyramid has a base area of 49 cm^2 and a total lateral area of 84 cm^2. What is the surface area?

A. 133 cm^2

B. 84 cm^2

C. 49 cm^2

D. 133 cm^3

Show your work:

Stretch Your Thinking
Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Two pyramids have the same base area (64 in^2) but different slant heights. Pyramid A has slant height 6 in and Pyramid B has slant height 10 in. How much more surface area does Pyramid B have? Why does slant height affect surface area but NOT base area?

Sentence starter: Pyramid A: SA = 64 + 4(½ × 8 × 6) = 64 + ____ = ____ in^2. Pyramid B: SA = 64 + 4(½ × 8 × 10) = 64 + ____ = ____ in^2. Pyramid B has ____ more in^2 because ____. Slant height only affects ____ because ____.

Show your work:

____

____

____

____

Reflect — Exit Ticket

A square pyramid has a base edge of 6 in and a slant height of 5 in. What is the total surface area?

A. 96 in^2
B. 60 in^2
C. 96 in^3
D. 36 in^2

Your answer:

____

____

____

Answer Key & Teacher Guide

1. **Try It 1:** A. 24 in^2 *— Area of a triangle = ½ × base × height = ½ × 8 × 6 = 24 in^2.*
2. **Try It 2:** A. 133 cm^2 *— SA = Base Area + Lateral Area = 49 + 84 = 133 cm^2.*
3. **Exit Ticket:** A. 96 in^2 *— Base: 6 × 6 = 36 in^2. Each lateral face: ½ × 6 × 5 = 15 in^2. Four faces: 4 × 15 = 60 in^2. SA = 36 + 60 = 96 in^2. Surface area uses square units (in^2).*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Pyramid is a solid with a flat bottom and triangle sides that meet at one point on top.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).